AI/ML INTERVIEW QUESTIONS & ANSWERS

Python • SQL • Statistics • Machine Learning • Deep Learning • NLP
Generative AI • LLMs • RAG • Agentic AI • MLOps • Projects

185 concise questions with answers across 20 topic areas

Beginner-friendly explanations designed for interview revision and as starter knowledge for a RAG interview-preparation chatbot. Answers are concise study notes; verify implementation-specific details against the versions and documentation you use.

How to use this guide

1. Read the question and try answering aloud before revealing the answer.
2. Ask the chatbot to explain unfamiliar terms with examples.
3. Practice follow-up and scenario questions, not only definitions.
4. For RAG, retain source metadata so the chatbot can cite document names and page numbers.

1. Python Programming

Q1. What is Python?

Answer: Python is a high-level, interpreted programming language known for readable syntax and a large ecosystem. It is widely used in automation, data analysis, machine learning, APIs, and AI applications.

Q2. What is the difference between a list, tuple, set, and dictionary?

Answer: A list is ordered and mutable; a tuple is ordered and immutable; a set stores unique elements; and a dictionary stores key-value pairs. Choose based on whether order, mutability, uniqueness, or key-based lookup matters.

Q3. What is the difference between mutable and immutable objects?

Answer: Mutable objects can be changed after creation, such as lists and dictionaries. Immutable objects cannot be changed in place, such as strings, integers, and tuples whose contents are themselves immutable.

Q4. What is the difference between == and is?

Answer: == compares values for equality. is checks whether two references point to the same object. Use is None to test for None, not equality as a substitute for identity.

Q5. What are *args and **kwargs?

Answer: *args collects extra positional arguments into a tuple; **kwargs collects extra keyword arguments into a dictionary. They help functions accept flexible inputs.

Q6. What is a list comprehension?

Answer: A list comprehension is a compact way to create a list from an iterable, optionally filtering values. Example: squares = [x*x for x in nums if x > 0].

Q7. What is a generator in Python?

Answer: A generator produces values lazily, often using yield. It can reduce memory use when processing large sequences because it does not need to store every result at once.

Q8. What is a decorator?

Answer: A decorator wraps a function or class to add behavior without changing its core implementation. Common uses include logging, timing, caching, and authorization.

Q9. What is exception handling?

Answer: Exception handling uses try and except to handle expected failures. else runs when no exception occurs, and finally runs whether or not an exception occurred, making it useful for cleanup.

Q10. What is OOP?

Answer: Object-oriented programming organizes code around objects that combine data and behavior. Four commonly taught principles are encapsulation, abstraction, inheritance, and polymorphism.

Q11. What is the difference between a shallow copy and a deep copy?

Answer: A shallow copy creates a new outer object but keeps references to nested objects. A deep copy recursively copies nested objects, which helps when independent nested data is needed.

Q12. What is a virtual environment?

Answer: A virtual environment isolates a project's Python packages from other projects. It reduces dependency conflicts and helps make setup reproducible.

Q13. What is the difference between multithreading and multiprocessing?

Answer: Threads share a process's memory and can help with I/O-bound work. Processes have separate memory and can use multiple CPU cores for CPU-bound work, though they have more overhead.

Q14. How do you remove duplicates from a list while preserving order?

Answer: For hashable values, use list(dict.fromkeys(items)). For complex or unhashable values, build an explicit loop with an appropriate equality or key function.

Q15. How do you count word frequency in Python?

Answer: Normalize the text, split it into tokens, and count with collections.Counter. For production NLP, consider punctuation, case, Unicode, and language-specific tokenization.

2. SQL and Databases

Q16. What is SQL?

Answer: SQL is a language for querying and managing relational databases. It supports selecting, inserting, updating, deleting, defining schemas, and controlling transactions and permissions.

Q17. What is the difference between WHERE and HAVING?

Answer: WHERE filters rows before grouping; HAVING filters groups after GROUP BY. Use WHERE for row-level conditions and HAVING for aggregate conditions such as COUNT(*) > 5.

Q18. What is the difference between INNER JOIN and LEFT JOIN?

Answer: INNER JOIN returns only matching rows from both tables. LEFT JOIN returns every row from the left table and matching right-side data, with NULLs when no match exists.

Q19. What is a primary key versus a foreign key?

Answer: A primary key uniquely identifies a row and cannot be NULL. A foreign key references a key in another or the same table and helps enforce referential integrity.

Q20. What is database normalization?

Answer: Normalization organizes relational data to reduce duplication and update anomalies. Common normal forms include 1NF, 2NF, and 3NF, each imposing progressively stronger structural rules.

Q21. What is an index?

Answer: An index is a data structure that can speed up reads and filtering, but it consumes storage and can slow inserts, updates, and deletes because the index must be maintained.

Q22. What are ACID properties?

Answer: Atomicity means all-or-nothing; Consistency preserves database rules; Isolation controls interactions between concurrent transactions; Durability means committed changes persist.

Q23. What is the difference between DELETE, TRUNCATE, and DROP?

Answer: DELETE removes selected rows and can use WHERE. TRUNCATE removes all rows in a table in database-specific ways. DROP removes the table object itself. Transaction and identity behavior varies by database.

Q24. What is a window function?

Answer: A window function computes a value across related rows without collapsing them into one row per group. Examples include ROW_NUMBER, RANK, running totals, and moving averages.

Q25. How do you find the second-highest salary?

Answer: One approach is SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees); this returns the second distinct salary if one exists.

Q26. What is SQL injection?

Answer: SQL injection happens when untrusted input is interpreted as SQL code. Use parameterized queries, least-privilege database accounts, and input validation; do not build SQL by concatenating raw user input.

3. Statistics and Mathematics

Q27. What is the difference between mean, median, and mode?

Answer: Mean is the arithmetic average, median is the middle value after sorting, and mode is the most frequent value. Median is often more robust to extreme outliers than the mean.

Q28. What is variance versus standard deviation?

Answer: Variance is the average squared deviation from the mean under a chosen population or sample definition. Standard deviation is its square root and is expressed in the original units.

Q29. What is the difference between covariance and correlation?

Answer: Covariance indicates whether two variables tend to move together but depends on their scales. Correlation standardizes the relationship to a bounded measure, usually between -1 and +1.

Q30. Why does correlation not imply causation?

Answer: Two variables can correlate because of confounding, reverse causality, selection effects, or coincidence. Establishing causation generally requires a suitable experimental or causal-inference design.

Q31. What is Bayes' theorem?

Answer: Bayes' theorem updates the probability of a hypothesis using evidence: $P(A|B) = P(B|A)P(A)/P(B)$, when $P(B)$ is nonzero.

Q32. What is a p-value?

Answer: A p-value is the probability, assuming the null hypothesis and model assumptions are true, of observing results at least as extreme as those obtained. It is not the probability that the null hypothesis is true.

Q33. What are Type I and Type II errors?

Answer: Type I error is rejecting a true null hypothesis (a false positive in many tests). Type II error is failing to reject a false null hypothesis (a false negative).

Q34. What is the Central Limit Theorem?

Answer: Under suitable conditions, the sampling distribution of a properly standardized sample mean approaches a normal distribution as sample size grows. It supports many statistical inference methods.

Q35. What is gradient descent?

Answer: Gradient descent updates model parameters in the direction that reduces a loss function, using the negative gradient. Learning rate controls step size; too large can diverge and too small can be slow.

Q36. What is entropy in a decision tree?

Answer: Entropy measures class uncertainty in a node. A split with greater information gain reduces entropy more, though implementations may also use Gini impurity.

4. Data Analysis and Preprocessing

Q37. What is EDA?

Answer: Exploratory Data Analysis is the process of inspecting data distributions, missing values, duplicates, outliers, relationships, and possible quality issues before modeling.

Q38. What is the difference between normalization and standardization?

Answer: Normalization often rescales values to a fixed range such as 0–1. Standardization commonly transforms values to have mean 0 and standard deviation 1. The exact term can vary by context.

Q39. How do you handle missing values?

Answer: First investigate why values are missing and how much is missing. Depending on the data, use deletion, median/mean imputation, a categorical missing label, model-based imputation, or missingness indicators. Fit imputation on training data only.

Q40. What is one-hot encoding?

Answer: One-hot encoding represents a categorical value as a vector of binary indicator columns. It avoids implying an artificial order but can create many columns for high-cardinality features.

Q41. What is data leakage?

Answer: Data leakage occurs when training uses information that would not legitimately be available at prediction time. Split data first and fit preprocessing, feature selection, and resampling only on training folds.

Q42. What is feature engineering?

Answer: Feature engineering creates or transforms input variables to better represent the problem, such as extracting month from a date or calculating ratios from valid source fields.

Q43. What is class imbalance?

Answer: Class imbalance means target classes occur at very different rates. Accuracy can be misleading; inspect precision, recall, F1, PR-AUC, and consider class weights or resampling within training data.

Q44. What is a Pandas DataFrame?

Answer: A DataFrame is a two-dimensional labeled table with rows and columns. Pandas supports filtering, grouping, joining, reshaping, cleaning, and reading data from formats such as CSV and Excel.

5. Machine Learning Fundamentals

Q45. What is machine learning?

Answer: Machine learning uses data to learn patterns for predictions or decisions rather than relying only on explicitly programmed rules. Its performance must be evaluated on data representative of intended use.

Q46. What is the difference between supervised and unsupervised learning?

Answer: Supervised learning uses labeled examples, such as inputs paired with outcomes. Unsupervised learning seeks structure in data without target labels, such as clusters or lower-dimensional representations.

Q47. What is the difference between classification and regression?

Answer: Classification predicts a discrete category, such as spam or not spam. Regression predicts a numerical value, such as house price or delivery time.

Q48. What is the difference between training, validation, and test data?

Answer: Training data fits model parameters; validation data supports model selection and tuning; test data estimates final performance. Repeatedly tuning against the test set makes it less independent.

Q49. What is overfitting?

Answer: Overfitting occurs when a model learns training-specific noise or patterns that do not generalize. Reduce it with suitable validation, simpler models, regularization, more representative data, or early stopping.

Q50. What is underfitting?

Answer: Underfitting occurs when a model is too simple or poorly trained to capture important patterns. It often performs poorly on both training and validation data.

Q51. What is the bias-variance trade-off?

Answer: High bias reflects overly restrictive assumptions; high variance reflects sensitivity to training data. Model selection balances these errors to improve generalization.

Q52. What is cross-validation?

Answer: Cross-validation repeatedly trains and evaluates a model on different splits to estimate generalization and compare choices. Keep preprocessing inside each fold to avoid leakage.

Q53. What is regularization?

Answer: Regularization adds a penalty to the training objective to discourage overly complex models. L1 can encourage sparse coefficients; L2 discourages large coefficients smoothly.

Q54. What is hyperparameter tuning?

Answer: Hyperparameter tuning selects settings such as tree depth or learning rate using validation or cross-validation. The final test set should not be used to repeatedly choose settings.

Q55. What is the difference between bagging and boosting?

Answer: Bagging trains models independently on varied samples and combines them, often reducing variance. Boosting builds models sequentially to focus on remaining errors and can reduce bias, but may be sensitive to noise.

Q56. What is data drift versus concept drift?

Answer: Data drift means the input distribution changes. Concept drift means the relationship between inputs and target changes. Both can degrade deployed model performance.

6. Regression Algorithms

Q57. What is linear regression?

Answer: Linear regression models a numeric target as a linear combination of features. It is often fitted by minimizing squared errors and relies on assumptions for particular forms of statistical inference.

Q58. What is the difference between linear and logistic regression?

Answer: Linear regression predicts a continuous numeric value, such as price. Logistic regression estimates class probabilities, commonly through a sigmoid for binary classification, and applies a threshold to produce a class label.

Q59. Why is logistic regression used for classification?

Answer: It maps a linear score through a logistic function to estimate a probability between 0 and 1. It is a relatively interpretable baseline for classification, although its decision boundary is linear in the features unless transformed.

Q60. What are MAE, MSE, and RMSE?

Answer: MAE averages absolute errors and is relatively easy to interpret. MSE averages squared errors and penalizes large errors more. RMSE is the square root of MSE and is in the target's units.

Q61. What is R-squared?

Answer: R-squared compares model residual variation with the variation around the target mean under a standard definition. It measures fit relative to a baseline and does not prove causation or guarantee good predictions on new data.

Q62. What is multicollinearity?

Answer: Multicollinearity occurs when predictors are strongly related, making coefficient estimates unstable or difficult to interpret. Check correlations or VIF and consider feature removal, regularization, or domain-informed alternatives.

Q63. What is Ridge versus Lasso regression?

Answer: Ridge uses an L2 penalty and shrinks coefficients toward zero. Lasso uses an L1 penalty and can set some coefficients exactly to zero. Elastic Net combines both penalties.

Q64. What are residuals?

Answer: Residuals are observed values minus predicted values. Plotting them can reveal nonlinearity, changing variance, outliers, and other model misspecification.

7. Classification and Evaluation Metrics

Q65. What is a confusion matrix?

Answer: A confusion matrix counts predicted versus actual classes. In binary classification it contains true positives, true negatives, false positives, and false negatives.

Q66. What is precision?

Answer: Precision = TP / (TP + FP). It answers: among cases predicted positive, what fraction were actually positive? It matters when false positives are costly.

Q67. What is recall?

Answer: Recall = TP / (TP + FN). It answers: among all actual positive cases, what fraction did the model find? It matters when missing positives is costly.

Q68. What is F1-score?

Answer: F1 is the harmonic mean of precision and recall: 2PR/(P+R), when the denominator is nonzero. It balances the two but does not include true negatives directly.

Q69. What is accuracy?

Answer: Accuracy is the fraction of predictions that are correct. It can be misleading when one class dominates or when different errors have different costs.

Q70. When is recall more important than precision?

Answer: Recall may be prioritized in screening or safety cases where missing a real positive is costly. The acceptable trade-off should be chosen with domain experts and the operating threshold.

Q71. What is ROC-AUC?

Answer: ROC-AUC measures how well scores rank positives above negatives across thresholds. It is useful for ranking, but PR-AUC can be more informative when positive cases are rare.

Q72. What is a classification report?

Answer: A classification report typically summarizes precision, recall, F1-score, and support for each class, plus averages. Always interpret it with class balance and the application context.

Q73. What are macro, micro, and weighted averages?

Answer: Macro averages metrics equally across classes. Weighted averages weight each class by its support. Micro aggregates counts across classes before computing the metric, which emphasizes overall instance-level performance.

Q74. Why can accuracy be misleading for imbalanced data?

Answer: If 95% of examples are negative, predicting every case as negative gives 95% accuracy but zero recall for positives. Use class-aware metrics and inspect the confusion matrix.

Q75. What is a decision threshold?

Answer: A threshold converts predicted probabilities or scores into class decisions. Lowering it often increases recall and false positives; raising it often increases precision while missing more positives, depending on the model and data.

8. Unsupervised Learning

Q76. What is clustering?

Answer: Clustering groups examples based on similarity without using predefined target labels. The resulting groups depend on feature representation, distance, algorithm, and parameter choices.

Q77. How does K-Means work?

Answer: K-Means initializes K centroids, assigns points to their nearest centroid, updates centroids to cluster means, and repeats until convergence or a stopping rule. It is sensitive to scale, initialization, and outliers.

Q78. How do you choose K in K-Means?

Answer: Use domain knowledge and diagnostics such as the elbow curve and silhouette score. These are aids rather than guarantees; check whether the clusters are stable and useful.

Q79. What is DBSCAN?

Answer: DBSCAN forms density-based clusters using neighborhood radius and minimum-point settings. It can find irregular shapes and label noise, but struggles when clusters have very different densities.

Q80. What is PCA?

Answer: Principal Component Analysis transforms correlated numeric features into orthogonal components ordered by explained variance. It can reduce dimensionality, but components may be harder to interpret.

Q81. What is the silhouette score?

Answer: The silhouette score compares how close a point is to its own cluster versus the nearest alternative cluster. Higher values generally indicate better-separated clusters, but the metric has limitations for complex shapes.

9. Deep Learning

Q82. What is deep learning?

Answer: Deep learning uses neural networks with multiple layers to learn representations. It is powerful for complex data such as images, speech, and text but often needs more data and compute than simpler methods.

Q83. What is an activation function?

Answer: An activation function introduces nonlinearity into a neural network. Common examples include ReLU for hidden layers, sigmoid for binary probability outputs, and softmax for multiclass probability outputs.

Q84. What is backpropagation?

Answer: Backpropagation applies the chain rule to compute gradients of a loss with respect to network parameters. An optimizer then uses those gradients to update weights.

Q85. What are epochs, batch size, and iterations?

Answer: An epoch is one pass through the training dataset. Batch size is the number of examples processed per update. An iteration usually refers to one parameter-update step.

Q86. What is dropout?

Answer: Dropout randomly disables a subset of activations during training to reduce co-adaptation and help regularize a neural network. It is typically disabled during inference.

Q87. What is a CNN?

Answer: A Convolutional Neural Network uses learned filters to detect local patterns, often sharing weights across positions. CNNs are widely used for images and other grid-like data.

Q88. What is an RNN?

Answer: A Recurrent Neural Network processes sequences while maintaining a hidden state. Basic RNNs can struggle with long dependencies due to vanishing or exploding gradients.

Q89. What are LSTM and GRU?

Answer: LSTM and GRU are gated recurrent architectures designed to manage information over longer sequences more effectively than a basic RNN.

Q90. What is transfer learning?

Answer: Transfer learning starts from a model trained on a related task and adapts it to a new task. It can reduce data and training requirements, especially when the source model is relevant.

Q91. What is the difference between TensorFlow and PyTorch?

Answer: Both are deep-learning frameworks. PyTorch is known for an imperative Python-first workflow; TensorFlow provides a broad ecosystem for training and deployment. The best choice depends on the project and team's tools.

10. NLP

Q92. What is NLP?

Answer: Natural Language Processing is the area of AI concerned with processing and generating human language, including classification, extraction, search, translation, and summarization.

Q93. What is tokenization?

Answer: Tokenization splits text into units such as words, subwords, or characters. LLMs commonly use subword tokens, so a token is not necessarily a whole word.

Q94. What is stemming versus lemmatization?

Answer: Stemming crudely strips word endings and may produce non-words. Lemmatization maps forms to dictionary lemmas using linguistic information and is often more linguistically accurate.

Q95. What is TF-IDF?

Answer: TF-IDF weights terms by their frequency in a document and how rare they are across the corpus. It is a useful lexical representation but does not inherently capture meaning or paraphrases.

Q96. What are word embeddings?

Answer: Embeddings represent words, sentences, or documents as numeric vectors intended to capture useful semantic or contextual relationships. Similarity measures can compare these vectors.

Q97. What is named entity recognition?

Answer: NER identifies spans of text that belong to categories such as people, organizations, dates, locations, or product names.

Q98. What is cosine similarity?

Answer: Cosine similarity compares the angle between two vectors. It is commonly used for embeddings, but its usefulness depends on the embedding model, normalization, and task.

11. AI Fundamentals

Q99. What is the difference between AI, ML, and deep learning?

Answer: AI is the broad field of building systems that perform tasks associated with intelligence. ML is a subset that learns patterns from data. Deep learning is a subset of ML based on multilayer neural networks.

Q100. What is an AI agent?

Answer: An AI agent observes its environment and chooses actions toward a goal. In LLM applications, an agent may use a model to select tools, inspect results, and decide what to do next.

Q101. What is the difference between BFS and DFS?

Answer: Breadth-first search explores nodes level by level and finds shortest paths in unweighted graphs when implemented correctly. Depth-first search explores a path deeply before backtracking and typically uses less frontier memory in some graph structures.

Q102. What is A* search?

Answer: A* ranks paths using f(n) = g(n) + h(n), where g is cost so far and h estimates remaining cost. With suitable conditions, such as an admissible heuristic for standard graph-search settings, it can find an optimal path.

Q103. What is reinforcement learning?

Answer: Reinforcement learning trains an agent to choose actions based on rewards and interaction with an environment. Core concepts include states, actions, policies, rewards, and value functions.

12. Generative AI and LLMs

Q104. What is Generative AI?

Answer: Generative AI creates new content such as text, images, audio, or code based on learned patterns. It can be useful for drafting and synthesis but may generate inaccurate or unsafe outputs.

Q105. What is an LLM?

Answer: A Large Language Model is a neural network trained on large text or multimodal datasets to model language. Many generate text by predicting token distributions conditioned on context.

Q106. How does an LLM generate text?

Answer: The model processes the prompt and produces a probability distribution for the next token. A decoding method selects a token, appends it to context, and repeats until a stopping condition.

Q107. What is a token?

Answer: A token is a unit used by a model's tokenizer. It may represent a word, part of a word, punctuation, or another symbol; token counts affect context limits and API cost.

Q108. What is a context window?

Answer: A context window is the maximum amount of tokenized input and output a model can process in one request, subject to the model's limits.

Q109. What is prompt engineering?

Answer: Prompt engineering structures instructions, context, examples, and output requirements to guide a model. Good prompts help but do not guarantee correctness or security.

Q110. What is zero-shot versus few-shot prompting?

Answer: Zero-shot prompting provides instructions without examples. Few-shot prompting includes a small number of examples to demonstrate the desired task or format.

Q111. What is hallucination?

Answer: A hallucination is an unsupported or false model-generated claim. Reduce risk with retrieval, clear constraints, source citations, validation, and abstention when evidence is insufficient.

Q112. What is temperature?

Answer: Temperature changes the sharpness of the next-token distribution. Lower values usually make outputs more predictable; higher values generally increase variety. Exact behavior depends on the model and decoding setup.

Q113. What is fine-tuning?

Answer: Fine-tuning continues training a pretrained model on task-specific examples. It can adapt style or behavior, but requires suitable data and evaluation and does not automatically provide up-to-date factual knowledge.

Q114. What is LoRA?

Answer: Low-Rank Adaptation trains small low-rank update matrices while keeping most pretrained weights frozen. It can reduce trainable parameters and memory requirements for adaptation.

Q115. What is function calling?

Answer: Function calling lets a model produce a structured request to invoke a defined tool. The application must validate arguments, authorize the action, execute it, and handle the result safely.

13. Embeddings and Vector Databases

Q116. What is an embedding?

Answer: An embedding is a numeric vector representing content such as text. Embedding models are trained so vector relationships are useful for tasks such as semantic search or clustering.

Q117. What is semantic search?

Answer: Semantic search retrieves content based on meaning represented by features or embeddings, helping match paraphrases that do not share exact words.

Q118. What is a vector database?

Answer: A vector database stores vectors and supports similarity search, often with metadata filtering and persistence. It is useful for retrieval systems such as RAG.

Q119. What are FAISS and ChromaDB?

Answer: FAISS is a library for efficient similarity search and clustering of dense vectors. ChromaDB is a vector database/tooling option that supports storing embeddings, documents, and metadata for retrieval applications.

Q120. What is hybrid search?

Answer: Hybrid search combines lexical matching, such as BM25, with semantic vector retrieval. It can improve coverage when exact terms, identifiers, and semantic paraphrases all matter.

Q121. What is reranking?

Answer: Reranking takes an initial candidate set and scores candidates more carefully against the query, often improving relevance at extra computational cost.

Q122. What is metadata filtering?

Answer: Metadata filtering restricts retrieval to chunks that match fields such as source, topic, date, user permissions, or difficulty level. Access-control filters must be enforced server-side.

14. Retrieval-Augmented Generation (RAG)

Q123. What is RAG?

Answer: Retrieval-Augmented Generation retrieves relevant external content and supplies it to a language model to generate a grounded response. It helps connect answers to a maintained knowledge base but does not guarantee correctness.

Q124. What are the main stages of a RAG pipeline?

Answer: A common pipeline has ingestion, text extraction, cleaning, chunking, embedding, indexing, query processing, retrieval, optional reranking, prompt construction, generation, and source citation.

Q125. What is the difference between RAG and fine-tuning?

Answer: RAG retrieves information at query time from an external knowledge base and can cite its sources. Fine-tuning changes model parameters using training examples. RAG is often preferable for changing factual documents; fine-tuning is often used to adapt behavior or task patterns.

Q126. What is document chunking?

Answer: Chunking divides long documents into smaller segments that can be embedded and retrieved. Good chunks preserve enough context to answer a question without becoming too broad or too fragmented.

Q127. What is chunk overlap?

Answer: Chunk overlap repeats some text between adjacent chunks to preserve context near boundaries. Too much overlap increases index size and duplicate retrieval; the right amount should be evaluated.

Q128. How do you choose chunk size?

Answer: Start with chunks suited to the document structure and embedding/model context limits. Compare retrieval and answer quality across sizes using a representative test set instead of assuming one size is universally best.

Q129. What is top-k retrieval?

Answer: Top-k retrieval returns the k highest-ranked candidates for a query. A small k may miss needed evidence; a large k may add noise and consume more context. Tune it empirically.

Q130. What is maximal marginal relevance (MMR)?

Answer: MMR balances relevance to the query with diversity among selected results. It can reduce near-duplicate chunks in the final context.

Q131. What is query rewriting?

Answer: Query rewriting transforms a user's question into a clearer or retrieval-friendly query. It can help with ambiguity and conversational references, but rewritten queries should preserve the user's intent.

Q132. How can a RAG chatbot cite sources?

Answer: Store metadata such as document name, page number, section, and chunk ID during ingestion. Return citations linked to the retrieved chunks and ensure the generated claims are actually supported by those sources.

Q133. How can RAG reduce hallucinations?

Answer: Provide relevant trusted evidence, instruct the model to answer only from that evidence, request citations, and allow it to abstain when context is insufficient. Evaluate failures because retrieval alone cannot eliminate hallucinations.

Q134. How do you evaluate RAG?

Answer: Evaluate retrieval quality with metrics such as recall@k and precision@k, and answer quality with correctness, relevance, faithfulness, and citation accuracy. Use a representative set of human-reviewed questions.

Q135. What is a RAG ingestion pipeline?

Answer: It loads source files, extracts text and metadata, cleans and chunks content, generates embeddings, and writes chunks to an index. It should be repeatable and handle updates and failures.

Q136. What is a conversational RAG chatbot?

Answer: Conversational RAG uses chat history to interpret follow-up questions, often rewriting them into standalone queries before retrieval. It should not blindly include all history because irrelevant turns can distract retrieval.

Q137. What if the answer is not in the PDF?

Answer: The chatbot should say the supplied documents do not contain enough information, ask a clarifying question if appropriate, or use an explicitly approved additional source. It should not invent a source citation.

Q138. How do you handle scanned PDFs?

Answer: Scanned PDFs may contain images rather than selectable text. Apply OCR, check extraction quality, retain page metadata, and evaluate tables or layouts that OCR may distort.

Q139. How do you update a RAG knowledge base?

Answer: Track source versions, reprocess changed documents, update or remove stale chunks, and verify the index after changes. Avoid mixing old and new versions unintentionally.

Q140. What is agentic RAG?

Answer: Agentic RAG uses an agent to decide whether and how to retrieve, rewrite queries, call tools, or perform multiple retrieval steps. It adds flexibility but also complexity, latency, and new failure modes.

15. Agentic AI

Q141. What is Agentic AI?

Answer: Agentic AI describes systems that use models to plan or choose actions toward a goal, often with tools and iterative feedback. It is more action-oriented than a single request-response chatbot.

Q142. What is the difference between a workflow and an agent?

Answer: A workflow follows a mostly predefined sequence of steps. An agent chooses some next actions dynamically based on context and tool results. Workflows are often easier to test and control.

Q143. What are the components of an AI agent?

Answer: Common components include a model, instructions, tools, state or memory, an action loop, observations, stopping conditions, and safety checks.

Q144. What is a multi-agent system?

Answer: A multi-agent system uses multiple agents or specialized roles that coordinate on a task. It can separate responsibilities but adds communication overhead and more complex debugging.

Q145. What is human-in-the-loop?

Answer: Human-in-the-loop means a person reviews or approves selected decisions or actions. It is especially important for high-impact, irreversible, or permission-sensitive operations.

Q146. How do you prevent an agent loop?

Answer: Set maximum steps, tool-call limits, timeouts, budgets, and explicit stopping criteria. Detect repeated tool calls and return a safe failure when progress stalls.

Q147. What is LangGraph?

Answer: LangGraph is a framework for defining stateful, graph-based LLM workflows and agents. It supports explicit control flow, state, checkpointing, and human approval patterns.

Q148. When should you not use an agent?

Answer: Avoid agents when a deterministic function or simple workflow solves the problem more reliably, cheaply, and transparently. Use autonomy only where dynamic decisions provide real value.

16. APIs, FastAPI, and Streamlit

Q149. What is FastAPI?

Answer: FastAPI is a Python framework for building APIs with type hints, request validation, and automatic OpenAPI documentation. It is commonly used to expose AI or ML functionality.

Q150. What is Streamlit?

Answer: Streamlit is a Python framework for building interactive data apps and prototypes without writing much frontend code. It can provide a chat UI for a RAG application.

Q151. Why separate the model backend and Streamlit frontend?

Answer: Separation makes the retrieval/model logic reusable, allows independent testing and deployment, keeps API secrets on the server, and lets different interfaces use the same backend.

Q152. What is a REST API?

Answer: A REST-style API exposes resources or operations through HTTP methods and predictable URLs, with standard status codes and request/response formats.

Q153. What is the difference between GET and POST?

Answer: GET is generally used to retrieve a resource or representation. POST commonly submits data or requests an operation that may create a resource or trigger processing.

Q154. What is Pydantic?

Answer: Pydantic validates and parses data using Python type annotations and models. FastAPI commonly uses it to validate request and response structures.

Q155. What is CORS?

Answer: Cross-Origin Resource Sharing is a browser security mechanism that controls whether a web page can request resources from another origin. Configure allowed origins carefully rather than allowing everything by default.

Q156. How do you secure API keys?

Answer: Store secrets in environment variables or a secrets manager, keep them out of source control and frontend code, restrict access, rotate them when needed, and avoid logging them.

17. MLOps and Deployment

Q157. What is MLOps?

Answer: MLOps applies engineering practices to the ML lifecycle, including reproducible training, versioning, deployment, monitoring, and controlled updates.

Q158. What is Docker?

Answer: Docker packages an application and its dependencies into an image that can run as a container. It helps standardize environments but does not automatically solve secrets, networking, or production security.

Q159. What is the difference between an image and a container?

Answer: An image is a packaged template. A container is a running or stopped instance created from an image.

Q160. What is model serialization?

Answer: Serialization saves model state or objects so they can be loaded later. Use trusted artifacts and compatible versions; Python pickle-based formats can execute code when loaded.

Q161. What is CI/CD?

Answer: Continuous Integration validates code changes through automated checks. Continuous Delivery or Deployment automates preparing or releasing changes to environments.

Q162. What is model monitoring?

Answer: Model monitoring tracks service health and, where possible, data distributions, prediction behavior, quality labels, latency, errors, and cost to detect issues after deployment.

Q163. What is batch inference versus online inference?

Answer: Batch inference predicts for many records together, often on a schedule. Online inference returns predictions in response to individual or small groups of live requests.

Q164. What is a rollback?

Answer: A rollback restores a previous known-good application or model version when a release causes failures or unacceptable quality.

18. Responsible AI and Security

Q165. What is bias in AI?

Answer: Bias can arise from unrepresentative data, historical inequalities, measurement choices, or model design. Assess it with context-appropriate subgroup metrics and mitigation methods.

Q166. What is prompt injection?

Answer: Prompt injection is an attempt to manipulate an LLM through untrusted instructions in a prompt or retrieved content. Treat retrieved text as data, separate trusted instructions, restrict tools, and validate outputs.

Q167. Why is an LLM output not automatically trusted?

Answer: LLM output may be incorrect, manipulated, malformed, or unsafe. Validate it against schemas, policies, permissions, and business rules before using it for consequential actions.

Q168. What is least privilege?

Answer: Least privilege gives a component only the permissions needed for its task. For agents, expose narrow tools and restrict data and actions by user authorization.

Q169. How should private documents be protected in RAG?

Answer: Apply authentication and document-level access control before retrieval, secure storage and transport, minimize logs, protect API keys, and verify that retrieved chunks are authorized for the requesting user.

19. Project and Scenario-Based Questions

Q170. How would you explain an AI/ML project in an interview?

Answer: Use a clear structure: problem, data, preprocessing, model choices, evaluation, deployment, your contribution, challenges, and next improvements. Be accurate about what you personally implemented.

Q171. How would you build a RAG chatbot for AI/ML interview preparation?

Answer: Collect trusted question-and-answer documents, extract text, preserve topic and page metadata, chunk content, embed and index it, retrieve relevant passages for a question, and generate an answer grounded in those passages. Add citations, evaluation, and an abstention rule.

Q172. Your model has high training accuracy but low test accuracy. What would you do?

Answer: Investigate overfitting, data leakage, split quality, distribution differences, and model complexity. Use cross-validation, regularization, simpler models, and more representative data as appropriate.

Q173. Your classifier has high recall but low precision. What does it mean?

Answer: It finds many actual positives but also flags many negatives as positive. Review the threshold, calibration, class costs, feature quality, and precision-recall trade-off.

Q174. Your RAG chatbot retrieves irrelevant chunks. How would you debug it?

Answer: Inspect extracted text and metadata first, then test chunk boundaries, embedding model, query formulation, similarity scores, metadata filters, top-k, and reranking. Use a fixed set of queries to compare changes.

Q175. Your RAG chatbot retrieves good context but answers incorrectly. What would you inspect?

Answer: Check whether the prompt clearly prioritizes evidence, whether context is too long or contradictory, whether citations support the answer, and whether the model is following untrusted instructions in the retrieved text.

Q176. Your RAG chatbot answers questions absent from the source. What should it do?

Answer: It should state that the knowledge base does not provide enough evidence, optionally ask for clarification, and avoid fabricating an answer or citation.

Q177. Your AI application has high latency. What would you measure?

Answer: Measure each stage separately: document retrieval, reranking, model generation, network time, and UI/backend overhead. Then optimize the actual bottleneck with caching, smaller context, efficient models, or async processing.

Q178. Why did you choose Random Forest for a tabular prediction project?

Answer: A reasonable answer is that Random Forest captures nonlinear relationships and feature interactions with limited feature scaling requirements. The choice should be justified by validation results against a baseline, not by the algorithm's reputation alone.

Q179. How do you evaluate an interview chatbot?

Answer: Create a reviewed set of questions with expected answer points and source references. Test retrieval relevance, factual correctness, completeness, citation accuracy, refusal when evidence is absent, and consistency across difficulty levels.

Q180. How do you answer when you do not know an interview question?

Answer: Be honest about the boundary of your knowledge, explain what you do understand, and describe how you would verify or investigate the topic. Do not bluff with fabricated technical claims.

20. HR and Communication

Q181. Tell me about yourself.

Answer: Give a 45–90 second summary of your education, relevant skills, one or two projects, and the type of role you are targeting. Connect your experience to the job description and avoid listing every tool.

Q182. Why do you want to work in AI/ML?

Answer: Explain what interests you about turning data into useful predictions or AI applications, then support it with a concrete project or learning experience.

Q183. Why should we hire you as a fresher?

Answer: Highlight foundations, evidence from projects, willingness to learn, and communication. Use specific examples rather than claiming expertise in every technology.

Q184. What is one technical weakness you are working on?

Answer: Choose a real but manageable gap, explain the concrete steps you are taking to improve it, and give an example of recent progress.

Q185. Describe a technical challenge you solved.

Answer: Use a concise situation-task-action-result structure. Explain your own actions, how you validated the solution, and what you learned.